

Source folder: /Users/jonat/Documents/GitHub/master_thesis
Files in this part: 17

File: results_old_broken/dp_4_alpha_0.5_dim_16_clients_100_seed_42.json

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  "loss": 0.2705486794392167,
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  "samples": 37,
  "epochs": 1
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  "samples": 46,
  "epochs": 1
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  "loss": 0.47698317960004344,
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  {
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    "epochs": 1
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    "client_id": "client_99",
    "loss": 0.3929808121298955,
    "samples": 76,
    "epochs": 1
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"timestamp": "2025-12-01T16:07:57.203674"
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"final_metrics": {
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  "mae": 0.4530077323162128,
  "accuracy": 0.5492468134414832,
  "samples": 1726,
  "NDCG@10": 0.0,
  "Hit@10": 0.06,
  "Precision@10": 0.006000000000000001,
  "Recall@10": 0.0009235280848184075,
  "num_users_evaluated": 50
},
"client_metrics": [],
"mobile_metrics": []
}

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File: results_old_broken/dp_4_alpha_0.5_dim_16_clients_100_seed_42_summary.csv

```

Round,Train_Loss,NDCG@10,Hit@10,Precision@10,Recall@10,MSE,MAE,Accuracy,Num_Clients>Total_Samples
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2562,0.5492468134414832,97,6706
3,0.46846008677903095,0.0,0.12,0.013999999999999999,0.005681665288116901,0.450759764411082,0.45152606499982
46,0.5492468134414832,97,6706
4,0.47437258753547634,0.0,0.06,0.006000000000000001,0.0011150234741784037,0.45072978866197005,0.45171751373
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9,0.47334652602097477,0.0,0.06,0.006000000000000001,0.0009235280848184075,0.45074231565205825,0.45273216132
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62128,0.5492468134414832,97,6706
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134414832,,

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File: results_old_broken/dp_4_alpha_0.5_dim_16_clients_100_seed_456.json

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{
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  "timestamp": "2025-12-01T17:52:14.008965",
  "config": {
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    "num_items": 4032,
    "embedding_dim": 16,
    "num_clients": 100,
    "alpha": 0.5,
    "dp_epsilon": 4,
    "use_dp": true,
    "dp_sigma": 5.668749999999999,
    "dp_clip_norm": 1.0,

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"num_rounds": 10,
"local_epochs": 1,
"learning_rate": 0.01,
"batch_size": 32,
"seed": 456
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"rounds": [
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  "test_metrics": {
    "mse": 0.4507654123483085,
    "mae": 0.45115810584358934,
    "accuracy": 0.5492468134414832,
    "samples": 1726,
    "NDCG@10": 0.0,
    "Hit@10": 0.14,
    "Precision@10": 0.014000000000000002,
    "Recall@10": 0.002751648588719527,
    "num_users_evaluated": 50
  },
  "aggregation": {
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    "round": 80,
    "num_clients": 97,
    "total_samples": 6706
  },
  "client_metrics": [
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    {
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      "loss": 0.6352470321155586,
      "samples": 53,
      "epochs": 1
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    {
      "client_id": "client_3",
      "loss": 0.6100999268734801,
      "samples": 92,
      "epochs": 1
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    {
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      "loss": 0.44072915928935097,
      "samples": 56,
      "epochs": 1
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    {
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      "loss": 0.4538710193893884,
      "samples": 81,
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    {
      "client_id": "client_6",
      "loss": 0.45439535011384075,
      "samples": 64,
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    {
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      "loss": 0.3857997363280155,
      "samples": 75,
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    {
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    "loss": 0.561990053879648,
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  {
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    "loss": 0.3991559863578349,
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  {
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    "loss": 0.6099434425487255,
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  {
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    "loss": 0.39055912589043745,
    "samples": 86,
    "epochs": 1
  },
  {
    "client_id": "client_15",
    "loss": 0.6512918089456685,
    "samples": 68,
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  {
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  {
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  "samples": 52,
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    "samples": 64,
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    "client_id": "client_93",
    "loss": 0.6433346929143338,
    "samples": 55,
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  {
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    "loss": 0.5154343317438539,
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    "loss": 0.42661115214531725,
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  "samples": 1726,
  "NDCG@10": 0.0,
  "Hit@10": 0.08,
  "Precision@10": 0.008,
  "Recall@10": 0.0009396472144948373,
  "num_users_evaluated": 50
},
"client_metrics": [],
"mobile_metrics": []
}
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File: results_old_broken/dp_4_alpha_0.5_dim_16_clients_100_seed_456_summary.csv

```
Round,Train_Loss,NDCG@10,Hit@10,Precision@10,Recall@10,MSE,MAE,Accuracy,Num_Clients,Total_Samples
1,0.46529210696521317,0.0,0.14,0.014000000000000002,0.002751648588719527,0.4507654123483085,0.4511581058435
8934,0.5492468134414832,97,6706
2,0.4740820488751516,0.0,0.06,0.006000000000000001,0.0026216984965461198,0.450768121769641,0.45133976698750
555,0.5492468134414832,97,6706
3,0.46908607171148226,0.0,0.08,0.008,0.003345713782784721,0.4507532999661955,0.45154480917567036,0.54924681
34414832,97,6706
4,0.4692630221500443,0.0,0.14,0.014000000000000002,0.007605898547285705,0.4507461297967895,0.45171041698610
52,0.5492468134414832,97,6706
5,0.47390549161135975,0.0,0.1,0.01,0.004592114440950085,0.45079621109735923,0.4519530160402229,0.5492468134
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6,0.4742810440288067,0.0,0.06,0.008,0.0009616860267130803,0.45077330240092717,0.4521412678140616,0.54924681
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7,0.474512799298417,0.0,0.1,0.01,0.004312147666484043,0.4507472644261109,0.45228901329349946,0.549246813441
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285,0.5492468134414832,97,6706
9,0.4656517578199143,0.0,0.06,0.008,0.0006803631812977607,0.4507273883836156,0.45261605727990284,0.54924681
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File: results_old_broken/dp_8_alpha_0.5_dim_16_clients_100_seed_123.json

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    "num_clients": 100,
    "alpha": 0.5,
    "dp_epsilon": 8,
    "use_dp": true,
    "dp_sigma": 3.11640625,
    "dp_clip_norm": 1.0,
    "num_rounds": 10,
    "local_epochs": 1,
    "learning_rate": 0.01,
    "batch_size": 32,
    "seed": 123
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  "rounds": [
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        "mae": 0.4509869097005727,
        "accuracy": 0.5492468134414832,
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        "Hit@10": 0.04,
        "Precision@10": 0.004,
        "Recall@10": 0.0010591133004926108,

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  "round": 40,
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  "total_samples": 6706
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  {
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    "loss": 0.6526512518623165,
    "samples": 53,
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    "loss": 0.6081798632231368,
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    "samples": 75,
    "epochs": 1
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    "loss": 0.3866850586608238,
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    "loss": 0.44404513463614675,
    "samples": 98,
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    "samples": 40,
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{
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  "samples": 71,
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  "loss": 0.2771358807967398,
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  "loss": 0.4428801053703787,
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        "samples": 55,
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        "loss": 0.3802084958899003,
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    {
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        "samples": 85,
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    "accuracy": 0.5492468134414832,
    "samples": 1726,
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    "Hit@10": 0.14,
    "Precision@10": 0.014000000000000002,
    "Recall@10": 0.0033667202693899024,
    "num_users_evaluated": 50
},
"client_metrics": [],
"mobile_metrics": []
}

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File: results_old_broken/dp_8_alpha_0.5_dim_16_clients_100_seed_123_summary.csv

Round,Train_Loss,NDCG@10,Hit@10,Precision@10,Recall@10,MSE,MAE,Accuracy,Num_Clients,Total_Samples
1,0.4698810865340609,0.0,0.04,0.004,0.0010591133004926108,0.4507595485773451,0.4509869097005727,0.549246813


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4414832, 97, 6706
2, 0.46468435671933145, 0.0, 0.08, 0.008, 0.0023256077256077253, 0.4507350817339998, 0.4510489913569127, 0.54924681
34414832, 97, 6706
3, 0.4678829217794078, 0.0, 0.04, 0.004, 0.0013636363636363635, 0.45073037531533355, 0.45112167062240016, 0.5492468
134414832, 97, 6706
4, 0.4699611341291024, 0.0, 0.08, 0.008, 0.0020919811371424273, 0.45071770715271475, 0.45115155057586637, 0.5492468
134414832, 97, 6706
5, 0.47449271536512794, 0.0, 0.1, 0.01, 0.0021798932250545154, 0.4507006427504096, 0.4512082516525463, 0.5492468134
414832, 97, 6706
6, 0.47296593070084286, 0.0, 0.08, 0.008, 0.0011552868184447132, 0.45070435129201014, 0.45126680924469736, 0.549246
8134414832, 97, 6706
7, 0.4684182188661335, 0.0, 0.16, 0.016, 0.004282599886930052, 0.45073215958693624, 0.45135772766961973, 0.54924681
34414832, 97, 6706
8, 0.4649116678108931, 0.0, 0.08, 0.01, 0.002647816812502721, 0.4506940128905472, 0.4514215984322381, 0.54924681344
14832, 97, 6706
9, 0.46792258259033975, 0.0, 0.12, 0.016, 0.0027422338896784188, 0.450653730994166, 0.45144933897263756, 0.54924681
34414832, 97, 6706
10, 0.4646940898326087, 0.0, 0.14, 0.014000000000000002, 0.0033667202693899024, 0.45065809884739916, 0.45152273100
745777, 0.5492468134414832, 97, 6706
FINAL,, 0.0, 0.14, 0.014000000000000002, 0.0033667202693899024, 0.45065809884739916, 0.45152273100745777, 0.549246
8134414832,,
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File: results_old_broken/dp_8_alpha_0.5_dim_16_clients_100_seed_42.json

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    "dp_epsilon": 8,
    "use_dp": true,
    "dp_sigma": 3.11640625,
    "dp_clip_norm": 1.0,
    "num_rounds": 10,
    "local_epochs": 1,
    "learning_rate": 0.01,
    "batch_size": 32,
    "seed": 42
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  "rounds": [
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        "mae": 0.45099866873695843,
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        "Precision@10": 0.004,
        "Recall@10": 0.0007838827838827838,
        "num_users_evaluated": 50
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          "samples": 60,
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          "client_id": "client_2",
          "loss": 0.6457272221593533,
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"samples": 53,
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  "loss": 0.6110315471607795,
  "samples": 92,
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  "loss": 0.4374444532662606,
  "samples": 56,
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  "loss": 0.4981842764587161,
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  "loss": 0.4599287069574344,
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  "epochs": 1
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},
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  "samples": 85,
  "epochs": 1
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      "samples": 55,
      "epochs": 1
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      "loss": 0.7382629994750676,
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      "epochs": 1
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      "samples": 70,
      "epochs": 1
    },
    {
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      "loss": 0.5156327739094965,
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      "samples": 85,
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    "Hit@10": 0.1,
    "Precision@10": 0.01,
    "Recall@10": 0.0031181485181485187,
    "num_users_evaluated": 50
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    "loss": 0.6289798750719063,
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    "loss": 0.44274638807619254,
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      "samples": 81,
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        "loss": 0.5347769862002331,
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    "mae": 0.4515894627930253,
    "accuracy": 0.5492468134414832,
    "samples": 1726,
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    "Hit@10": 0.12,
    "Precision@10": 0.014000000000000002,
    "Recall@10": 0.002152270375582251,
    "num_users_evaluated": 50
},
"client_metrics": [],
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File: results_old_broken/dp_8_alpha_0.5_dim_16_clients_100_seed_42_summary.csv

Round	Train_Loss	NDCG@10	Hit@10	Precision@10	Recall@10	MSE	MAE	Accuracy	Num_Clients	Total_Samples
1	0.4698897989052896	0.0	0.04	0.004	0.0007838827838827838	0.4507670179441105	0.45099866873695843	0.5492468134414832	97	6706
2	0.47446653321516247	0.0	0.1	0.01	0.0031181485181485187	0.4507570659713701	0.45105416197351345	0.5492468134414832	97	6706
3	0.4684081065856218	0.0	0.08	0.008	0.0011941465941465942	0.45075143668217843	0.45111460812812887	0.5492468134414832	97	6706
4	0.47580457123681513	0.0	0.1	0.01	0.007175999439157334	0.45072942854769676	0.4511811166898124	0.5492468134414832	97	6706
5	0.46396593523420354	0.0	0.12	0.014000000000000002	0.00832535885167464	0.4507342542447193	0.45124976797976885	0.5492468134414832	97	6706
6	0.4719003444530718	0.0	0.06	0.006000000000000001	0.0035202492211838005	0.4507357617656712	0.4512817751905457	0.5492468134414832	97	6706
7	0.4750572915841528	0.0	0.1	0.01	0.005589382448537379	0.4507753812424139	0.45136249604120177	0.5492468134414832	97	6706
8	0.4684505631579702	0.0	0.04	0.006000000000000001	0.0003099928109273904	0.45078330762416713	0.4514608316874587	0.5492468134414832	97	6706
9	0.47403069142686705	0.0	0.2	0.022000000000000002	0.009882825233158645	0.45078037318321396	0.4515325727495921	0.5492468134414832	97	6706
10	0.4708928460207671	0.0	0.12	0.014000000000000002	0.002152270375582251	0.4507695783192976	0.4515894627930			

253,0.5492468134414832,97,6706

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File: results_old_broken/dp_8_alpha_0.5_dim_16_clients_100_seed_456.json

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      "samples": 53,
      "epochs": 1
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    {
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      "loss": 0.6073602996306631,
      "samples": 92,
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    {
      "client_id": "client_4",
      "loss": 0.45833853812084624,
      "samples": 56,
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    {
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      "loss": 0.4535296079825104,
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    {
      "client_id": "client_7",
      "loss": 0.3606454223269989,
      "samples": 75,
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    {
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      "loss": 0.3982204954172482,
      "samples": 122,
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  ],
  {
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  "samples": 36,
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  "loss": 0.4105043875110033,
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  "loss": 0.3543871800972965,
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  "samples": 67,
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  "loss": 0.3754524797485068,
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  "samples": 55,
  "epochs": 1
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  "samples": 37,
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  "epochs": 1
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    {
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        "loss": 0.5045130855972566,
        "samples": 85,
        "epochs": 1
    },
    {
        "client_id": "client_99",
        "loss": 0.4082147608457562,
        "samples": 76,
        "epochs": 1
    }
],
"timestamp": "2025-12-01T14:24:04.119236"
}
],
"final_metrics": {
    "mse": 0.4506838055778627,
    "mae": 0.45155008477380076,
    "accuracy": 0.5492468134414832,
    "samples": 1726,
    "NDCG@10": 0.0,
    "Hit@10": 0.04,
    "Precision@10": 0.006000000000000001,
    "Recall@10": 0.0004047670639219935,
    "num_users_evaluated": 50
},
"client_metrics": [],
"mobile_metrics": []
}
```

File: results_old_broken/dp_8_alpha_0.5_dim_16_clients_100_seed_456_summary.csv

```
Round,Train_Loss,NDCG@10,Hit@10,Precision@10,Recall@10,MSE,MAE,Accuracy,Num_Clients>Total_Samples
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135,0.5492468134414832,97,6706
2,0.46308359117853365,0.0,0.1,0.01,0.002293352280618834,0.45074570489786175,0.45105536870856966,0.549246813
4414832,97,6706
3,0.46816034138061374,0.0,0.1,0.01,0.0018387441978991277,0.4507395725261295,0.4511229315058243,0.5492468134
414832,97,6706
4,0.47350490129183126,0.0,0.14,0.013999999999999999,0.0032387957592410117,0.45073350169291226,0.45118721998
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5,0.47112001425049493,0.0,0.08,0.01,0.0020444072056975283,0.4506861778841649,0.4512290490460976,0.549246813
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6,0.47632966023035683,0.0,0.1,0.01,0.0014926700539603765,0.4506764655041391,0.4513076780016425,0.5492468134
414832,97,6706
7,0.47821041305844286,0.0,0.08,0.01,0.0010984048209854661,0.45068553487229596,0.4513774896137684,0.54924681
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8,0.47355328978645284,0.0,0.06,0.006000000000000001,0.0007758241758241759,0.4506892193095847,0.451418383787
2944,0.5492468134414832,97,6706
9,0.47284733123086886,0.0,0.02,0.002,6.153846153846154e-05,0.4506670591590855,0.451495281100135,0.549246813
4414832,97,6706
10,0.47516674762204525,0.0,0.04,0.006000000000000001,0.0004047670639219935,0.4506838055778627,0.45155008477
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FINAL,,0.0,0.04,0.006000000000000001,0.0004047670639219935,0.4506838055778627,0.45155008477380076,0.5492468
134414832,,
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File: results_old_broken/dp_inf_alpha_0.1_dim_16_clients_100_seed_123.json

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  "timestamp": "2025-12-02T07:41:13.840214",
  "config": {
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    "num_items": 4032,
    "embedding_dim": 16,
    "num_clients": 100,
    "alpha": 0.1,
    "dp_epsilon": null,
    "use_dp": false,
    "dp_sigma": 1.0,
  }
}
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"dp_clip_norm": 1.0,
"num_rounds": 10,
"local_epochs": 1,
"learning_rate": 0.01,
"batch_size": 32,
"seed": 123
},
"rounds": [
  {
    "round": 1,
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    "test_metrics": {
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      "mae": 0.4509284935572194,
      "accuracy": 0.5492468134414832,
      "samples": 1726,
      "NDCG@10": 0.0,
      "Hit@10": 0.08,
      "Precision@10": 0.008,
      "Recall@10": 0.0025704284326533346,
      "num_users_evaluated": 50
    },
    "aggregation": {
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      "round": 160,
      "num_clients": 97,
      "total_samples": 6706
    },
    "client_metrics": [
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        "loss": 0.6422202433215033,
        "samples": 95,
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        "loss": 0.600934085786314,
        "samples": 54,
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      },
      {
        "client_id": "client_2",
        "loss": 0.5356109746658199,
        "samples": 28,
        "epochs": 1
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      {
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        "loss": 0.44796005853965554,
        "samples": 80,
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      {
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        "loss": 0.3436389784423897,
        "samples": 40,
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      {
        "client_id": "client_5",
        "loss": 0.4136133504056827,
        "samples": 85,
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      {
        "client_id": "client_6",
        "loss": 0.45458888252163693,
        "samples": 11,
        "epochs": 1
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      {
        "client_id": "client_7",
        "loss": 0.6869234286349195,
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{
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{
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    "samples": 1726,
    "NDCG@10": 0.0,
    "Hit@10": 0.08,
    "Precision@10": 0.008,
    "Recall@10": 0.0025704284326533346,
    "num_users_evaluated": 50
  },
  "client_metrics": [],
  "mobile_metrics": []
}
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File: results_old_broken/dp_inf_alpha_0.1_dim_16_clients_100_seed_123_summary.csv

```
Round,Train_Loss,NDCG@10,Hit@10,Precision@10,Recall@10,MSE,MAE,Accuracy,Num_Clients,Total_Samples
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134414832,97,6706
2,0.4737208675081942,0.0,0.08,0.008,0.0025704284326533346,0.4507412753679799,0.45092848692680704,0.54924681
34414832,97,6706
3,0.48078738295677226,0.0,0.08,0.008,0.0025704284326533346,0.4507412698426363,0.45092847698118843,0.5492468
134414832,97,6706
4,0.4734666963321777,0.0,0.08,0.008,0.0025704284326533346,0.4507412686685007,0.45092848029639465,0.54924681
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6,0.4774746594613852,0.0,0.08,0.008,0.0025704284326533346,0.45074126535329456,0.45092846372036366,0.5492468
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7,0.4750133630284135,0.0,0.08,0.008,0.0025704284326533346,0.4507412708095714,0.4509284570899513,0.549246813
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File: results_old_broken/dp_inf_alpha_0.1_dim_16_clients_100_seed_42.json

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    "num_clients": 100,
    "alpha": 0.1,
    "dp_epsilon": null,
    "use_dp": false,
    "dp_sigma": 1.0,
    "dp_clip_norm": 1.0,
    "num_rounds": 10,
    "local_epochs": 1,
    "learning_rate": 0.01,
    "batch_size": 32,
    "seed": 42
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  "rounds": [
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        "mae": 0.450935927354585,
        "accuracy": 0.5492468134414832,
        "samples": 1726,
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        "Hit@10": 0.06,
        "Precision@10": 0.0060000000000000001,

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    "samples": 54,
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    "loss": 0.535938740279674,
    "samples": 28,
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    "loss": 0.4270709620900595,
    "samples": 80,
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    "loss": 0.48431355877419746,
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    "loss": 0.4545862301175968,
    "samples": 11,
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  {
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    "loss": 0.6898827679660257,
    "samples": 59,
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    "loss": 0.2187207812212172,
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  {
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    "samples": 218,
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{
  "client_id": "client_61",
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{
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  "epochs": 1
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{
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  "samples": 74,
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  "loss": 0.3875619321402833,
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  "epochs": 1
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  "samples": 176,
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  "samples": 80,
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  "loss": 0.3443981773244556,
  "samples": 59,
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{
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    "loss": 0.47222428254458937,
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  {
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    "loss": 0.3878792820864022,
    "samples": 73,
    "epochs": 1
  },
  {
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    "loss": 0.5571928551168266,
    "samples": 89,
    "epochs": 1
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    "loss": 0.7513420103363038,
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    "epochs": 1
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    "loss": 0.5730069356469089,
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    "epochs": 1
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    "loss": 0.7058088413574817,
    "samples": 89,
    "epochs": 1
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    "loss": 0.7207331345692951,
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    "epochs": 1
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    "samples": 184,
    "epochs": 1
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  {
    "client_id": "client_99",
    "loss": 0.38913221628909067,
    "samples": 171,
    "epochs": 1
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"timestamp": "2025-12-02T07:40:59.391789"
}
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  "mae": 0.450935894202523,
  "accuracy": 0.5492468134414832,
  "samples": 1726,
  "NDCG@10": 0.0,
  "Hit@10": 0.06,
  "Precision@10": 0.006000000000000001,
  "Recall@10": 0.0016604544506983532,
  "num_users_evaluated": 50
},
"client_metrics": [],
"mobile_metrics": []
}
```

File: results_old_broken/dp_inf_alpha_0.1_dim_16_clients_100_seed_42_summary.csv

```
Round,Train_Loss,NDCG@10,Hit@10,Precision@10,Recall@10,MSE,MAE,Accuracy,Num_Clients>Total_Samples
1,0.4790909791297951,0.0,0.06,0.006000000000000001,0.0016604544506983532,0.4507463260846232,0.4509359273545
85,0.5492468134414832,97,6706
2,0.46922297213561637,0.0,0.06,0.006000000000000001,0.0016604544506983532,0.4507463293998294,0.450935927354
585,0.5492468134414832,97,6706
3,0.47880534690353194,0.0,0.06,0.006000000000000001,0.0016604544506983532,0.45074632829476063,0.45093592735
4585,0.5492468134414832,97,6706
4,0.470277929931708,0.0,0.06,0.006000000000000001,0.0016604544506983532,0.45074633160996685,0.4509359317748
599,0.5492468134414832,97,6706
5,0.4784515649996073,0.0,0.06,0.006000000000000001,0.0016604544506983532,0.45074632829476063,0.450935920724
1726,0.5492468134414832,97,6706
6,0.48021283606701937,0.0,0.06,0.006000000000000001,0.0016604544506983532,0.45074632159528144,0.45093591298
869146,0.5492468134414832,97,6706
7,0.47312308126868013,0.0,0.06,0.006000000000000001,0.0016604544506983532,0.45074632049021274,0.45093591298
869146,0.5492468134414832,97,6706
8,0.4776293411064556,0.0,0.06,0.006000000000000001,0.0016604544506983532,0.450746319385144,0.45093590193800
415,0.5492468134414832,97,6706
9,0.47686077100541013,0.0,0.06,0.006000000000000001,0.0016604544506983532,0.45074631054459413,0.45093590856
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10,0.47417071802386596,0.0,0.06,0.006000000000000001,0.0016604544506983532,0.450746316000871,0.450935894202
523,0.5492468134414832,97,6706
FINAL,,0.0,0.06,0.006000000000000001,0.0016604544506983532,0.450746316000871,0.450935894202523,0.5492468134
414832,,
```

File: results_old_broken/dp_inf_alpha_0.1_dim_16_clients_100_seed_456.json

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  "timestamp": "2025-12-02T09:25:00.619809",
  "config": {
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    "num_items": 4032,
    "embedding_dim": 16,
    "num_clients": 100,
    "alpha": 0.1,
    "dp_epsilon": null,
    "use_dp": false,
    "dp_sigma": 1.0,
    "dp_clip_norm": 1.0,
    "num_rounds": 10,
    "local_epochs": 1,
    "learning_rate": 0.01,
    "batch_size": 32,
    "seed": 456
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  "rounds": [
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      "train_loss": 0.48290893744739843,
      "test_metrics": {
        "mse": 0.4507665924235816,
        "mae": 0.4509478245245623,
        "accuracy": 0.5492468134414832,
        "samples": 1726,
        "NDCG@10": 0.0,
        "Hit@10": 0.1,
        "Precision@10": 0.01,
        "Recall@10": 0.0017506972559604137,
        "num_users_evaluated": 50
      },
      "aggregation": {
        "status": "aggregated",
        "round": 170,
        "num_clients": 97,
        "total_samples": 6706
      },
      "client_metrics": [
        {
          "client_id": "client_0",
          "loss": 0.6425408367306708,
          "samples": 95,
          "epochs": 1
        }
      ]
    }
  ]
}
```

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},
{
  "client_id": "client_1",
  "loss": 0.6150853538686084,
  "samples": 54,
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{
  "client_id": "client_2",
  "loss": 0.5357183821282976,
  "samples": 28,
  "epochs": 1
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{
  "client_id": "client_3",
  "loss": 0.42709974118978555,
  "samples": 80,
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  "client_id": "client_4",
  "loss": 0.4841882549723703,
  "samples": 40,
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  "loss": 0.40821791989087625,
  "samples": 85,
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},
{
  "client_id": "client_6",
  "loss": 0.4544550404256359,
  "samples": 11,
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{
  "client_id": "client_7",
  "loss": 0.6927082949365172,
  "samples": 59,
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  "client_id": "client_8",
  "loss": 0.18759254672286876,
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  "loss": 0.38191912934150246,
  "samples": 218,
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{
  "client_id": "client_11",
  "loss": 0.3223593113335432,
  "samples": 43,
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  "loss": 0.4929660504937587,
  "samples": 67,
  "epochs": 1
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  }
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  "loss": 0.47738215300013137,
  "samples": 43,
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  "samples": 120,
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  }
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  "samples": 42,
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  "loss": 0.4681788779089129,
  "samples": 113,
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{
  "client_id": "client_76",
  "loss": 0.33336076616899163,
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  "loss": 0.31252777286355554,
  "samples": 136,
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  "loss": 0.7344284557800043,
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},
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  "client_id": "client_81",
  "loss": 0.5712847361226431,
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{
  "client_id": "client_82",
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  "samples": 119,
  "epochs": 1
},
{
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},
{
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  "loss": 0.3558763172107171,
  "samples": 81,
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},
{
  "client_id": "client_86",
  "loss": 0.49792591027795174,
  "samples": 74,
  "epochs": 1
},
{
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  "loss": 0.318805214802505,
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},
{
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  "samples": 176,
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{
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  "loss": 0.5208028607046189,
  "samples": 80,
  "epochs": 1
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{

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    "client_id": "client_90",
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  {
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    "loss": 0.4495124953517845,
    "samples": 45,
    "epochs": 1
  },
  {
    "client_id": "client_92",
    "loss": 0.38780053444279566,
    "samples": 73,
    "epochs": 1
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  {
    "client_id": "client_93",
    "loss": 0.5455531149533898,
    "samples": 89,
    "epochs": 1
  },
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    "loss": 0.6825238033876039,
    "samples": 45,
    "epochs": 1
  },
  {
    "client_id": "client_95",
    "loss": 0.5810674876233984,
    "samples": 82,
    "epochs": 1
  },
  {
    "client_id": "client_96",
    "loss": 0.7089010696930415,
    "samples": 89,
    "epochs": 1
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    "loss": 0.7268904269457206,
    "samples": 119,
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  {
    "client_id": "client_98",
    "loss": 0.3715207702608485,
    "samples": 184,
    "epochs": 1
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  {
    "client_id": "client_99",
    "loss": 0.3991125951608199,
    "samples": 171,
    "epochs": 1
  }
],
"timestamp": "2025-12-02T09:35:23.000053"
},
{
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    "mse": 0.4507665913185128,
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    "accuracy": 0.5492468134414832,
    "samples": 1726,
    "NDCG@10": 0.0,
    "Hit@10": 0.1,
    "Precision@10": 0.01,
    "Recall@10": 0.0017506972559604137,
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"num_users_evaluated": 50
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  "round": 171,
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  "total_samples": 6706
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    "samples": 54,
    "epochs": 1
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    "loss": 0.5357183225237124,
    "samples": 28,
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File: results_old_broken/dp_inf_alpha 0.1 dim 16 clients 100_seed_456_summary.csv

Round	Train_Loss	NDCG@10	Hit@10	Precision@10	Recall@10	MSE	MAE	Accuracy	Num_Clients	Total_Samples
1	0.48290893744739843	0.0	0.1	0.01	0.0017506972559604137	0.4507665924235816	0.4509478245245623	0.5492468134414832	97	6706
2	0.4722284742751342	0.0	0.1	0.01	0.0017506972559604137	0.4507665913185128	0.45094782010428736	0.5492468134414832	97	6706
3	0.4744449313923754	0.0	0.1	0.01	0.0017506972559604137	0.45076659905399397	0.4509478223144248	0.5492468134414832	97	6706
4	0.4790825672008022	0.0	0.1	0.01	0.0017506972559604137	0.4507665868982379	0.45094781899921865	0.5492468134414832	97	6706
5	0.47074128416041155	0.0	0.1	0.01	0.0017506972559604137	0.45076659456465223	0.4509478178941499	0.5492468134414832	97	6706
6	0.47647819418597487	0.0	0.1	0.01	0.0017506972559604137	0.45076659463371904	0.45094781015866875	0.5492468134414832	97	6706
7	0.4753509828495835	0.0	0.1	0.01	0.0017506972559604137	0.45076659794892526	0.45094781015866875	0.5492468134414832	97	6706
8	0.47669684662156375	0.0	0.1	0.01	0.0017506972559604137	0.45076659345958353	0.4509478079485313	0.5492468134414832	97	6706

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4414832, 97, 6706
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4414832, 97, 6706
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File: results_old_broken/dp_inf_alpha_0.5_dim_16_clients_100_seed_123.json

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